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SIT225: Data Capture Technologies

Activity 6.1: Plotly data dashboard

Plotly Dash apps give a point-&-click interface to models written in Python, vastly expanding the notion of what's possible in a traditional "dashboard". With Dash apps, data scientists and engineers put complex Python analytics in the hands of business decision-makers and operators. In this activity, you will learn basic building blocks of Plotly to create Dash apps.

Hardware Required

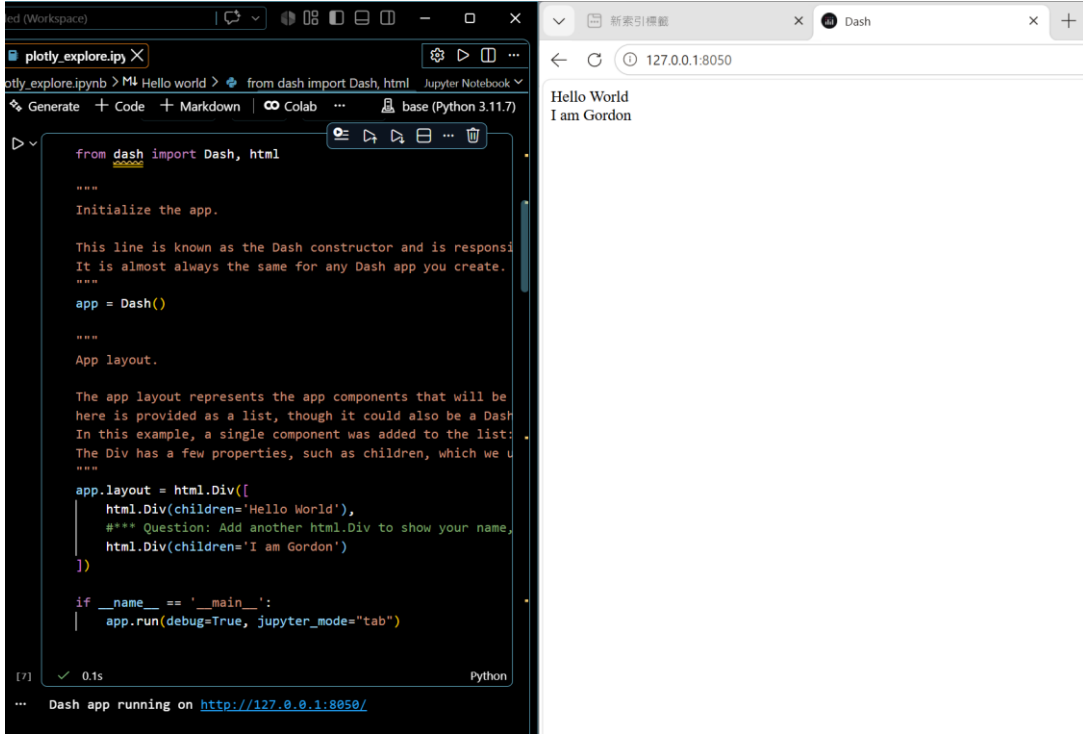
No hardware is required.

Software Required

Plotly library and Dash module
Python 3

Steps

Step	Action
1	Install Plotly and dash using the command below in the command line. <pre>\$ pip install plotly dash</pre> You can download Jupyter Notebook from here (https://github.com/deakin-deep-dreamer/sit225/blob/main/week_6/plotly_explore.ipynb) and run all the cells. The Notebook contains multiple sections such as Hello World which follows a sample code in a following cell. If you run the Hello world cell it will show Plotly Dash web page. The cell also includes a Question (**** Question) which you will need to carry out to get a modified output. You will need to capture the output and share the screenshot in the following steps.

2	Question: Hello World cell has a question - add another html.Div to show your name, and re-run the cell for output. You will need to update the code, run the cell, capture the screenshot of the output and paste it here. Answer:  The screenshot shows a Jupyter Notebook on the left and a web browser on the right. The notebook cell contains Python code for a Dash application. The code imports Dash and html, initializes the app, and sets the layout to a Div containing two other Divs: one with 'Hello World' and another with 'I am Gordon'. The browser on the right shows the output of the app at the URL 127.0.0.1:8050, displaying 'Hello World' and 'I am Gordon'.
3	Question: Connecting to Data cell has a question - change page size and observe the change in widget controls such as, total number of pages. You will need to update the code, run the cell, capture the screenshot of the output and paste it here. Answer:

```
Explorer (Ctrl+Shift+I) nb
C:\Users\gordo> OneDrive> Desktop> Y3> T2> SIT225 Data Capture> SIT225 2026T2> Web Jupyter Notebook
Generate Code Markdown Colab Run All Run All Restart base (Python 3.11.7)

Connecting to Data

There are many ways to add data to an app: APIs, external databases, local .txt files, JSON files, and more. In this example, we will highlight one of the most common ways of incorporating data from a CSV sheet.

# Import packages
# We import the dash_table module to display the data inside a Dash DataTable.
from dash import Dash, html, dash_table

# We also import the pandas library to read the CSV sheet data.
import pandas as pd

# Incorporate data
df = pd.read_csv('gapminder2007.csv')

# Explore data
print(df.head())
print("Data rowsXcols:", df.shape)

# Initialize the app
app = Dash()

# App layout.
# The 2nd item is a table with only 10 rows per page.
app.layout = html.Div([
    html.Div(children='My First App with Data'),
    dash_table.DataTable(data=df.to_dict('records'), page_size=10)
])

# Run the app
if __name__ == '__main__':
    app.run(debug=True, jupyter_mode='tab')
```

My First App with Data

country	pop	continent	lifeExp	gdpPerCap
Afghanistan	31889923	Asia	43.828	974.5803384
Albania	3600523	Europe	76.423	5937.629525999999
Algeria	33333216	Africa	72.301	6223.367465
Angola	12428476	Africa	42.731	4797.231267
Argentina	40301927	Americas	75.32	12779.37964
Australia	20434176	Oceania	81.235	34435.367439999995
Austria	8199783	Europe	79.829	36126.4927
Bahrain	708573	Asia	75.635	29796.04834
Bangladesh	150448339	Asia	64.062	1391.253792
Belgium	10392226	Europe	79.441	33692.60508

```
plotly_explore.ipynb
SIT225 2026T2> Web 6> plotly_explore.ipynb> M4 Connecting to Data> Import packages Jupyter Notebook
Generate Code Markdown Colab Run All Run All Restart base (Python 3.11.7)

# Import packages
# We import the dash_table module to display the data inside a Dash DataTable.
from dash import Dash, html, dash_table

# We also import the pandas library to read the CSV sheet data.
import pandas as pd

# Incorporate data
df = pd.read_csv('gapminder2007.csv')

# Explore data
print(df.head())
print("Data rowsXcols:", df.shape)

# Initialize the app
app = Dash()

# App layout.
# The 2nd item is a table with only 10 rows per page.
app.layout = html.Div([
    html.Div(children='My First App with Data'),
    dash_table.DataTable(data=df.to_dict('records'), page_size=5)
])

# Run the app
if __name__ == '__main__':
    app.run(debug=True, jupyter_mode='tab')
```

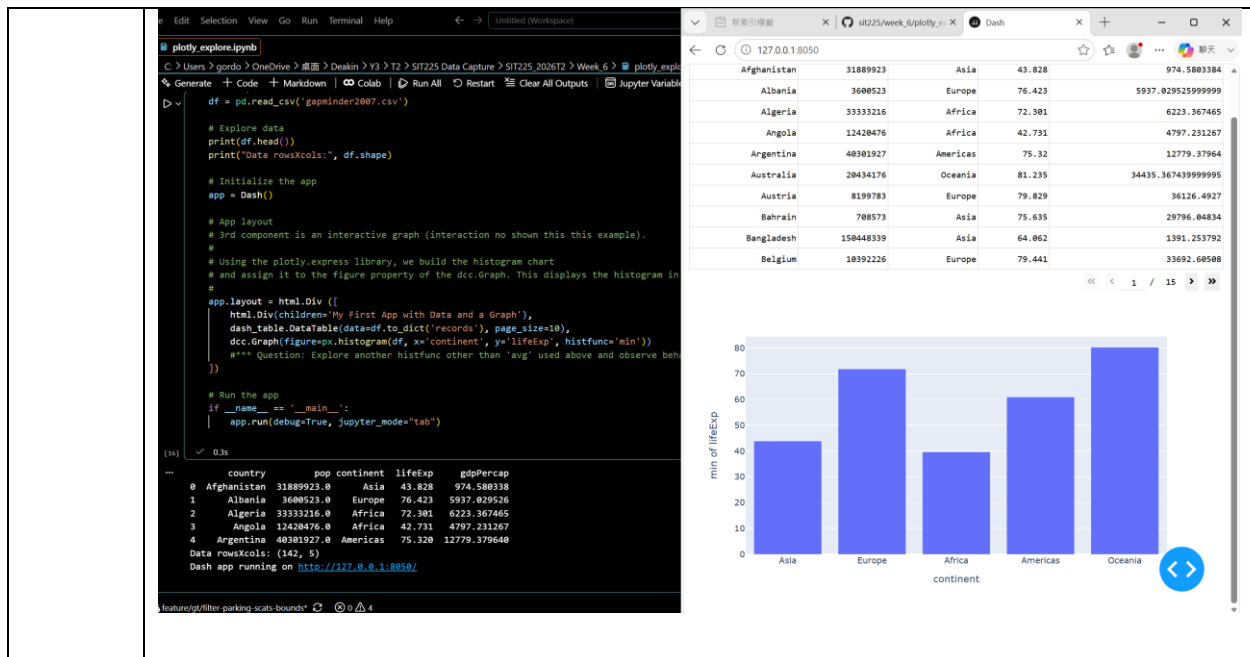
My First App with Data

country	pop	continent	lifeExp	gdpPerCap
Afghanistan	31889923	Asia	43.828	974.5803384
Albania	3600523	Europe	76.423	5937.629525999999
Algeria	33333216	Africa	72.301	6223.367465
Angola	12428476	Africa	42.731	4797.231267
Argentina	40301927	Americas	75.32	12779.37964

4

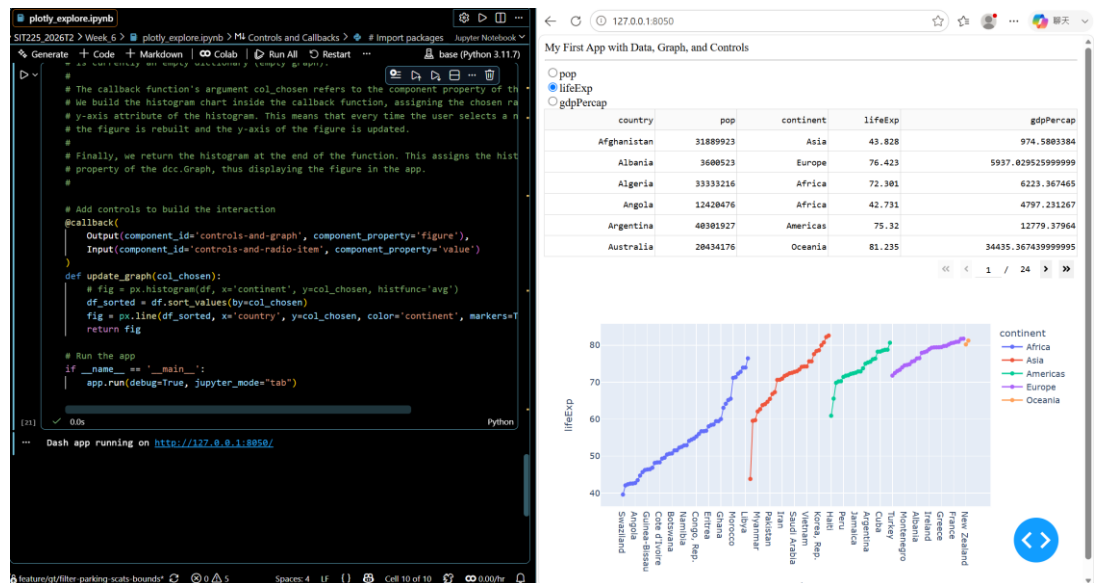
Question: Visualising data cell has a question - explore another histfunc other than 'avg' used above and observe behaviour. You will need to update the code, run the cell, capture the screenshot of the output and paste it here.

Answer:



5 **Question:** Controls and Callbacks cell has a question - use line graphs instead of histogram. You will need to update the code, run the cell, capture the screenshot of the output and paste it here.

Answer:



6 **Question:** Now you have learned how to use Plotly Dash for visualising your data, describe how you will be using this tool for your desired sensor

	monitoring dashboard with a number of sensors including DHT22 or accelerometer data. Answer: plotly dash is a tool to visualize data, since it manage by code base, I can use capture sensors data by Arduino cloud or python script then pass that data to dash so I can see the data in real time. And find out more patten.
7	Question: Convert the Notebook to PDF and merge with this activity sheet PDF. You will need this merged PDF to combine with this week's OnTrack task for submission.

```
# Fill in student ID and name
#
student_id = "s223532749"
student_first_last_name = "Gordon Tse"
print(student_id, student_first_last_name)

s223532749 Gordon Tse

# install plotly and dash, if not yet already
# ! pip install plotly dash

import plotly, dash
print(plotly.__version__)
print(dash.__version__)

5.9.0
2.14.2
```

Hello world

Building and launching an app with Dash can be done with just 5 lines of code. Follow the tutorial (<https://dash.plotly.com/tutorial>) for more detail.

```
from dash import Dash, html

"""
Initialize the app.

This line is known as the Dash constructor and is responsible for
initializing your app.
It is almost always the same for any Dash app you create.
"""
app = Dash()

"""
App layout.

The app layout represents the app components that will be displayed in
the web browser and
here is provided as a list, though it could also be a Dash component.
In this example, a single component was added to the list: an
html.Div.
The Div has a few properties, such as children, which we use to add
text content to the page: "Hello World".
"""
app.layout = html.Div([
    html.Div(children='Hello World'),
    *** Question: Add another html.Div to show your name, and re-run
```

```

the cell for output.
    html.Div(children='I am Gordon')
])

if __name__ == '__main__':
    app.run(debug=True, jupyter_mode="tab")

Dash app running on http://127.0.0.1:8050/
<IPython.core.display.Javascript object>

```

Connecting to Data

There are many ways to add data to an app: APIs, external databases, local .txt files, JSON files, and more. In this example, we will highlight one of the most common ways of incorporating data from a CSV sheet.

```

# Import packages

# We import the dash_table module to display the data inside a Dash
DataTable.
from dash import Dash, html, dash_table

# We also import the pandas library to read the CSV sheet data.
import pandas as pd

# Incorporate data
df = pd.read_csv('gapminder2007.csv')

# Explore data
print(df.head())
print("Data rowsXcols:", df.shape)

# Initialize the app
app = Dash()

# App layout.
# The 2nd item is a table with only 10 rows per page.
app.layout = html.Div([
    html.Div(children='My First App with Data'),
    dash_table.DataTable(data=df.to_dict('records'), page_size=5)
    *** Question: Change page size and observe the change in widget
    controls
    # such as, total number of pages.
])

```

```
# Run the app
if __name__ == '__main__':
    app.run(debug=True, jupyter_mode="tab")
```

	country	pop	continent	lifeExp	gdpPercap
0	Afghanistan	31889923.0	Asia	43.828	974.580338
1	Albania	3600523.0	Europe	76.423	5937.029526
2	Algeria	33333216.0	Africa	72.301	6223.367465
3	Angola	12420476.0	Africa	42.731	4797.231267
4	Argentina	40301927.0	Americas	75.320	12779.379640

Data rowsXcols: (142, 5)
Dash app running on http://127.0.0.1:8050/
<IPython.core.display.Javascript object>

Visualising data

The Plotly graphing library has more than 50 chart types to choose from. In this example, we will make use of the histogram chart.

```
# Import packages

# We import the dcc module (DCC stands for Dash Core Components).
# This module includes a Graph component called dcc.Graph, which is
# used to render interactive graphs.
from dash import Dash, html, dash_table, dcc

# We also import the plotly.express library to build the interactive
# graphs.
import plotly.express as px

import pandas as pd

# Incorporate data
df = pd.read_csv('gapminder2007.csv')

# Explore data
print(df.head())
print("Data rowsXcols:", df.shape)

# Initialize the app
app = Dash()

# App layout
# 3rd component is an interactive graph (interaction not shown in this
# example).
#
# Using the plotly.express library, we build the histogram chart
```

```

# and assign it to the figure property of the dcc.Graph. This displays
the histogram in our app.
#
app.layout = html.Div ([
    html.Div(children='My First App with Data and a Graph'),
    dash_table.DataTable(data=df.to_dict('records'), page_size=10),
    dcc.Graph(figure=px.histogram(df, x='continent', y='lifeExp',
histfunc='min'))
    *** Question: Explore another histfunc other than 'avg' used
above and observe behaviour.
])

# Run the app
if __name__ == '__main__':
    app.run(debug=True, jupyter_mode="tab")

```

	country	pop	continent	lifeExp	gdpPercap
0	Afghanistan	31889923.0	Asia	43.828	974.580338
1	Albania	3600523.0	Europe	76.423	5937.029526
2	Algeria	33333216.0	Africa	72.301	6223.367465
3	Angola	12420476.0	Africa	42.731	4797.231267
4	Argentina	40301927.0	Americas	75.320	12779.379640

Data rowsXcols: (142, 5)
Dash app running on http://127.0.0.1:8050/
<IPython.core.display.Javascript object>

Controls and Callbacks

So far you have built a static app that displays tabular data and a graph. However, as you develop more sophisticated Dash apps, you will likely want to give the app user more freedom to interact with the app and explore the data in greater depth. To achieve that, you will need to add controls to the app by using the callback function.

In this example we will add radio buttons to the app layout. Then, we will build the callback to create the interaction between the radio buttons and the histogram chart.

```

# Import packages

# We import dcc like we did in the previous section to use dcc.Graph.
# In this example, we need dcc for dcc.Graph as well as the radio
buttons component, dcc.RadioItems.
#
# To work with the callback in a Dash app, we import the callback
module and the two arguments
# commonly used within the callback: Output and Input.
#
from dash import Dash, html, dash_table, dcc, callback, Output, Input

```

```

import pandas as pd
import plotly.express as px

# Incorporate data
df = pd.read_csv('gapminder2007.csv')

# Initialize the app
app = Dash()

# App layout
app.layout = html.Div([
    html.Div(children='My First App with Data, Graph, and Controls'),
    html.Hr(),
    dcc.RadioItems(options=['pop', 'lifeExp', 'gdpPercap'],
value='lifeExp', id='controls-and-radio-item'),
    dash_table.DataTable(data=df.to_dict('records'), page_size=6),
    dcc.Graph(figure={}, id='controls-and-graph')
    *** Question: Use line graphs instead of histogram.
])

#
# Notice that we add that RadioItems component to the layout, directly
# above the DataTable.
# There are three options, one for every radio button.
# The lifeExp option is assigned to the value property, making it the
# currently selected value.
#
# Both the RadioItems and the Graph components were given id names:
# these will be used by
# the callback to identify the components.
#
# The inputs and outputs of our app are the properties of a particular
# component.
# In this example, our input is the value property of the component
# that has the ID "controls-and-radio-item".
# If you look back at the layout, you will see that this is currently
# lifeExp.
# Our output is the figure property of the component with the ID
# "controls-and-graph", which
# is currently an empty dictionary (empty graph).
#
# The callback function's argument col_chosen refers to the component
# property of the input lifeExp.
# We build the histogram chart inside the callback function, assigning
# the chosen radio item to the
# y-axis attribute of the histogram. This means that every time the
# user selects a new radio item,
# the figure is rebuilt and the y-axis of the figure is updated.
#
# Finally, we return the histogram at the end of the function. This

```

```

assigns the histogram to the figure
# property of the dcc.Graph, thus displaying the figure in the app.
#

# Add controls to build the interaction
@callback(
    Output(component_id='controls-and-graph',
            component_property='figure'),
    Input(component_id='controls-and-radio-item',
           component_property='value')
)
def update_graph(col_chosen):
    # fig = px.histogram(df, x='continent', y=col_chosen,
histfunc='avg')
    df_sorted = df.sort_values(by=col_chosen)
    fig = px.line(df_sorted, x='country', y=col_chosen,
                  color='continent', markers=True)
    return fig

# Run the app
if __name__ == '__main__':
    app.run(debug=True, jupyter_mode="tab")

Dash app running on http://127.0.0.1:8050/
<IPython.core.display.Javascript object>

```

Student name: Gordon Tse

Student ID:s223532749

SIT225: Data Capture Technologies

Activity 7.1: Data analysis and interpretation

Data analysis is a broad term that covers a wide range of techniques that enable you to reveal any insights and relationships that may exist within raw data. As you might expect, Python lends itself readily to data analysis. Once Python has analyzed your data, you can then use your findings to make good business decisions, improve procedures, and even make informed predictions based on what you've discovered.

You have done data wrangling using Python Pandas module already in activity 5.2. In this activity, you will learn Data science statistics and linear regression models.

Hardware Required

No hardware is required.

Software Required

Python 3

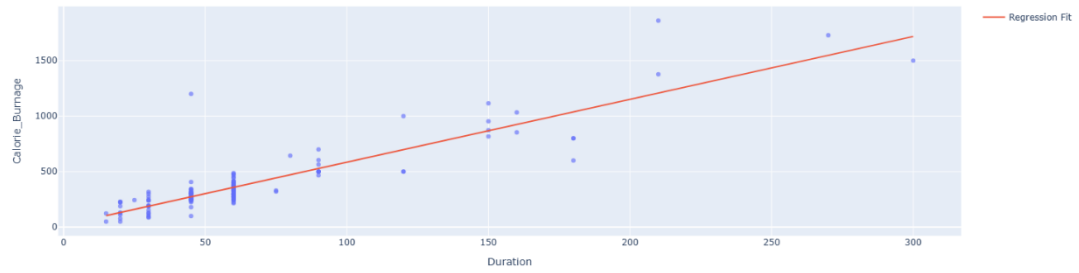
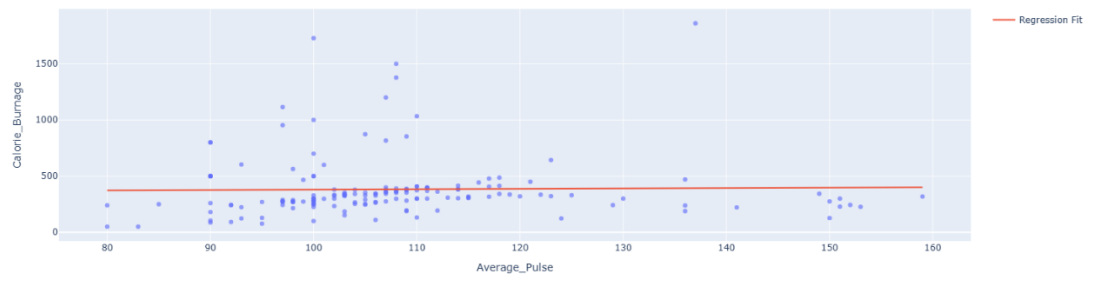
Python packages including Pandas, Numpy, Scikit-learn, seaborn, plotly

Steps:

Step	Action
1	A Jupyter Notebook is provided for Data Science exploration here (https://github.com/deakin-deep-dreamer/sit225/tree/main/week_7). You will need to fill in your student ID and name and run all the cells to observe the output. Convert the Notebook into PDF and merge with this activity sheet which needs to be combined with this week's task for OnTrack submission.

	Question: There are sections in the Notebook. After running the cells and observing the outputs, provide your reflection in brief on the topic items for each section of the Notebook. Answer: 1. Descriptive Statistics – this section I learn about different math statistics like sum, avg, count and percentile. 1.1 Percentile – it let me find the what is the number at different %. 1.2 Standard Deviation – it showing me the core topic of Statistics, this show how spread are the data. Coefficient of variation show how large is the sdt is. 1.3 Variance - multiply the standard deviation by itself = Variance. It also a way to show how spread are the data. 1.4 Correlation – to show the relationship between two variables. It show if x change, what will happened to y. 1 = perfect linear relationship: x increase y increase. 0 = no linear relationship: x increase y no change. -1 = perfect negative linear relationship: x increase y decrease. 1.5 Correlation Matrix – a array to show correlation. 1.6 Correlation Does not imply Causality – what I understand is when correlation cannot be apply to the variables. Like they have no relationship But getting Correlation = 1 or -1. 1.7 Linear Regression – it help to draw a straight line to show the relationship data. It can use for predict.
2	Question: In the 1.1 Percentile subsection of Descriptive statistics section in the Notebook, you have calculated 10%, 25%, 50% and 75% percentiles for <i>Max_Pulse</i>. Compare these percentiles with <i>Average_Pulse</i> percentiles for any trend, if exists. Answer: now I get the percentile for max_pulse, I found that overall in max_pulse is more then average_pulse which is normal because it showing the max number. also both when percentile increase the number also increase .

	<pre> avg_pulse = full_health_data["Average_Pulse"] print("percentile_10", np.percentile(avg_pulse, 10)) print("percentile_25", np.percentile(avg_pulse, 25)) print("percentile_50", np.percentile(avg_pulse, 50)) print("percentile_75", np.percentile(avg_pulse, 75)) ✓ 0.0s percentile_10 92.2 percentile_25 100.0 percentile_50 105.0 percentile_75 111.0 max_pulse = full_health_data["Max_Pulse"] print("percentile_10", np.percentile(max_pulse, 10)) print("percentile_25", np.percentile(max_pulse, 25)) print("percentile_50", np.percentile(max_pulse, 50)) print("percentile_75", np.percentile(max_pulse, 75)) ✓ 0.0s percentile_10 120.0 percentile_25 124.0 percentile_50 131.0 percentile_75 141.0 </pre>
3	Question: In the “Correlation Does not imply Causality” section answer the question regarding the increase of ice cream sale in your own understanding. Answer: Correlation is the measures the relation between two variables, when we talk about during summer both sale of ice cream and drowning accidents increase, I will say it kind of having Correlation because of warm weather. The weather is the cause. But since when weather getting hot, people will buy ice cream and go to swimming, so I will they have the correlation. In short the direct cause of both event is warm weather.
4	Question: In the 1.7 Linear Regression section in the Notebook, a linear regression model was used to predict Calorie_Burnage from attributes such as Average_Pulse. The Duration value was predicted from the model for all the value range of Average_Pulse and a regression line was drawn. You will need to answer the follow up question next to 1.7 section where it is required to generate a linear regression model for Duration instead of Average_Pulse to predict the Calorie_Burnage. Take a screenshot of the regression line and paste it here. Also, comment on both the regression lines. Answer: Looking at the Calorie_Burnage – Duration graph, is showing a upward trend and Calorie_Burnage - Average_Pulse is showing a flat line.





SIT225: Data Analysis & interpretation

Run each cell to generate output and finally convert this notebook to PDF.

```
In [1]: # Fill in student ID and name
#
student_id = "s223532749"
student_first_last_name = "Gordon Tse"
print(student_id, student_first_last_name)
```

s223532749 Gordon Tse

1. Descriptive Statistics

Descriptive statistics summarizes important features of a data set such as:

- Count
- Sum
- Standard deviation
- Percentile
- Average

```
In [2]: # Make sure necessary packages are already installed.
# !pip install pandas numpy seaborn

import pandas as pd
import numpy as np
import seaborn as sns

full_health_data = pd.read_csv("full_health_data.csv", header=0, sep=",")
print (full_health_data.describe())
```

	Duration	Average_Pulse	Max_Pulse	Calorie_Burnage	Hours_Work \
count	163.000000	163.000000	163.000000	163.000000	163.000000
mean	64.263804	107.723926	134.226994	382.368098	4.386503
std	42.994520	14.625062	16.403967	274.227106	3.923772
min	15.000000	80.000000	100.000000	50.000000	0.000000
25%	45.000000	100.000000	124.000000	256.500000	0.000000
50%	60.000000	105.000000	131.000000	320.000000	5.000000
75%	60.000000	111.000000	141.000000	388.500000	8.000000
max	300.000000	159.000000	184.000000	1860.000000	11.000000

	Hours_Sleep
count	163.000000
mean	7.680982
std	0.663934
min	5.000000
25%	7.500000
50%	8.000000
75%	8.000000
max	12.000000

1.1 Percentile

25%, 50% and 75% - Percentiles

Observe the output of the above cell for 25%, 50% and 75% of all the columns.

Let's explain for Average_Pulse:

- 25% of all of the training sessions have an average pulse of 100 beats per minute or lower. If we flip the statement, it means that 75% of all of the training sessions have an average pulse of 100 beats per minute or higher.
- 75% of all the training session have an average pulse of 111 or lower. If we flip the statement, it means that 25% of all of the training sessions have an average pulse of 111 beats per minute or higher.

```
In [3]: avg_pulse = full_health_data["Average_Pulse"]
print("percentile_10", np.percentile(avg_pulse, 10) )
print("percentile_25", np.percentile(avg_pulse, 25) )
print("percentile_50", np.percentile(avg_pulse, 50) )
print("percentile_75", np.percentile(avg_pulse, 75) )
```

```
percentile_10 92.2
percentile_25 100.0
percentile_50 105.0
percentile_75 111.0
```

Question: Calculate percentiles for Max_Pulse.

You should answer a follow up question in the activity sheet.

```
In [4]: max_pulse = full_health_data["Max_Pulse"]
print("percentile_10", np.percentile(max_pulse, 10) )
print("percentile_25", np.percentile(max_pulse, 25) )
print("percentile_50", np.percentile(max_pulse, 50) )
print("percentile_75", np.percentile(max_pulse, 75) )
```

```
percentile_10 120.0
percentile_25 124.0
percentile_50 131.0
percentile_75 141.0
```

1.2 Standard Deviation

Standard deviation is a number that describes how spread out the observations are.

A mathematical function will have difficulties in predicting precise values, if the observations are "spread". Standard deviation is a measure of uncertainty.

A low standard deviation means that most of the numbers are close to the mean (average) value.

A high standard deviation means that the values are spread out over a wider range.

```
In [5]: import numpy as np

# We can use the std() function from Numpy to find the standard deviation of a

std = np.std(full_health_data)
print(std)
```

```
Duration          42.862432
Average_Pulse     14.580131
Max_Pulse         16.353571
Calorie_Burnage   273.384624
Hours_Work        3.911718
Hours_Sleep       0.661895
dtype: float64
```

```
c:\Users\gordo\anaconda3\Lib\site-packages\numpy\core\fromnumeric.py:3643: FutureWarning: The behavior of DataFrame.std with axis=None is deprecated, in a future version this will reduce over both axes and return a scalar. To retain the old behavior, pass axis=0 (or do not pass axis)
    return std(axis=axis, dtype=dtype, out=out, ddof=ddof, **kwargs)
```

1.2.1 Coefficient of variation

In the above cell, what does standard deviation numbers mean?

The coefficient of variation is used to get an idea of how large the standard deviation is.

Mathematically, the coefficient of variation is defined as: $\text{Coefficient of Variation} = \text{Standard Deviation} / \text{Mean}$

```
In [6]: cv = np.std(full_health_data) / np.mean(full_health_data)
print(cv)

# We see that the variables Duration and Calorie_Burnage has
# a high Standard Deviation compared to Max_Pulse, Average_Pulse and Hours_Sleep
#
```

```
Duration          0.367051
Average_Pulse     0.124857
Max_Pulse         0.140043
Calorie_Burnage   2.341122
Hours_Work        0.033498
Hours_Sleep       0.005668
dtype: float64
```

1.3 Variance

Variance is another number that indicates how spread out the values are.

In fact, if you take the square root of the variance, you get the standard deviation. Or the other way around, if you multiply the standard deviation by itself, you get the variance!

```
In [7]: var = np.var(full_health_data)
print(var)
```

```
Duration          1837.188076
Average_Pulse     212.580225
Max_Pulse         267.439271
Calorie_Burnage   74739.152847
Hours_Work        15.301536
Hours_Sleep       0.438105
dtype: float64
```

```
c:\Users\gordo\anaconda3\Lib\site-packages\numpy\core\fromnumeric.py:3785: FutureWarning: The behavior of DataFrame.var with axis=None is deprecated, in a future version this will reduce over both axes and return a scalar. To retain the old behavior, pass axis=0 (or do not pass axis)
  return var(axis=axis, dtype=dtype, out=out, ddof=ddof, **kwargs)
```

1.4 Correlation

Correlation measures the relationship between two variables.

A function has a purpose to predict a value, by converting input (x) to output (f(x)). We can say also that a function uses the relationship between two variables for prediction.

Correlation Coefficient

The correlation coefficient measures the relationship between two variables.

The correlation coefficient can never be less than -1 or higher than 1.

- 1 = there is a perfect linear relationship between the variables
- 0 = there is no linear relationship between the variables
- -1 = there is a perfect negative linear relationship between the variables

Perfect Linear Relationship (Correlation Coefficient = 1)

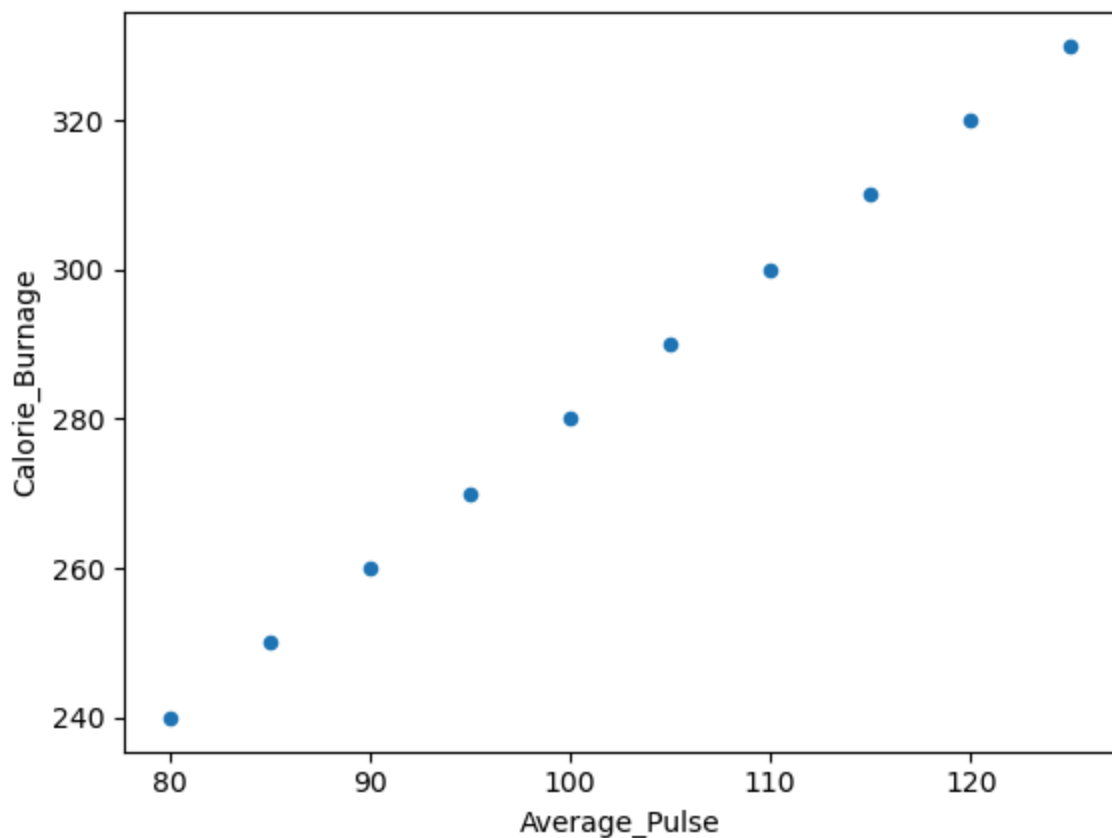
it exists a perfect linear relationship between Average_Pulse and Calorie_Burnage.

```
In [8]: # Positive correlation
#

import matplotlib.pyplot as plt

def create_linear_health_data():
    data = [
        {'Duration':30, 'Average_Pulse':80, 'Max_Pulse':120, 'Calorie_Burnage':100},
        {'Duration':45, 'Average_Pulse':85, 'Max_Pulse':120, 'Calorie_Burnage':150},
        {'Duration':45, 'Average_Pulse':90, 'Max_Pulse':130, 'Calorie_Burnage':200},
        {'Duration':60, 'Average_Pulse':95, 'Max_Pulse':130, 'Calorie_Burnage':250},
        {'Duration':60, 'Average_Pulse':100, 'Max_Pulse':140, 'Calorie_Burnage':300},
        {'Duration':60, 'Average_Pulse':105, 'Max_Pulse':140, 'Calorie_Burnage':350},
        {'Duration':60, 'Average_Pulse':110, 'Max_Pulse':145, 'Calorie_Burnage':400},
        {'Duration':45, 'Average_Pulse':115, 'Max_Pulse':145, 'Calorie_Burnage':450},
        {'Duration':60, 'Average_Pulse':120, 'Max_Pulse':150, 'Calorie_Burnage':500},
        {'Duration':45, 'Average_Pulse':125, 'Max_Pulse':150, 'Calorie_Burnage':550}
    ]
    return data

health_data = pd.DataFrame.from_dict(create_linear_health_data())
health_data.plot(x='Average_Pulse', y='Calorie_Burnage', kind='scatter')
plt.show()
```



Perfect Negative Linear Relationship (Correlation Coefficient = -1)

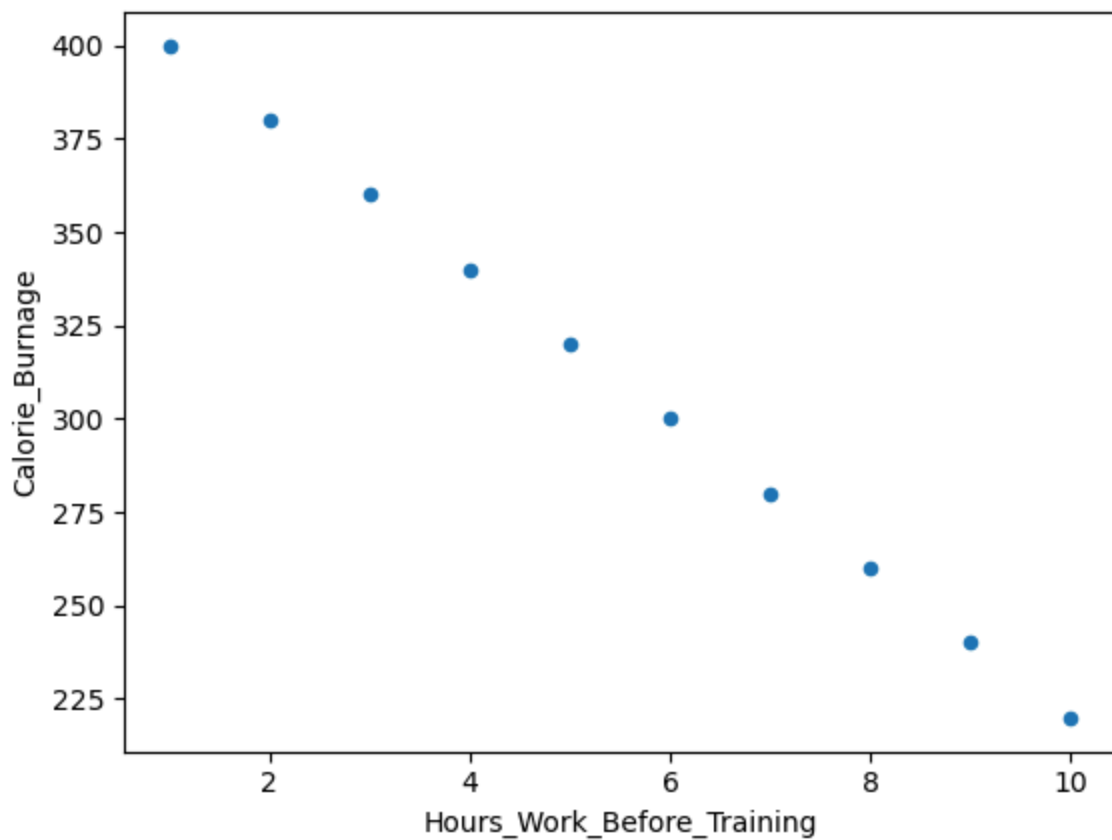
We have plotted fictional data here. The x-axis represents the amount of hours worked at our job before a training session. The y-axis is Calorie_Burnage.

If we work longer hours, we tend to have lower calorie burnage because we are exhausted before the training session.

The correlation coefficient here is -1.

```
In [9]: # Negative correlation
#
negative_corr = {'Hours_Work_Before_Training': [10,9,8,7,6,5,4,3,2,1],
                 'Calorie_Burnage': [220,240,260,280,300,320,340,360,380,400]}
negative_corr = pd.DataFrame(data=negative_corr)

negative_corr.plot(x='Hours_Work_Before_Training', y='Calorie_Burnage', kind=
plt.show())
```

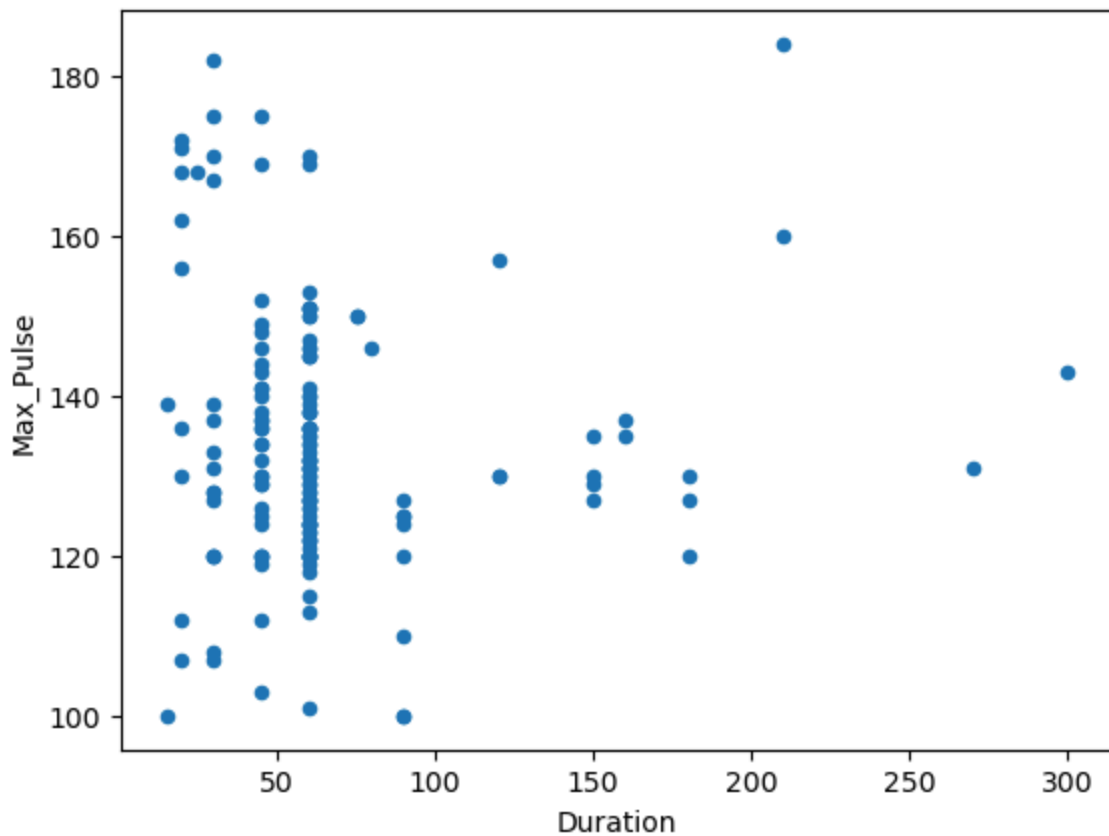


No Linear Relationship (Correlation coefficient = 0)

As you can see, there is no linear relationship between the two variables. It means that longer training session does not lead to higher Max_Pulse.

The correlation coefficient here is 0.

```
In [10]: full_health_data.plot(x='Duration', y='Max_Pulse', kind='scatter')
plt.show()
```



1.5 Correlation Matrix

A matrix is an array of numbers arranged in rows and columns.

A correlation matrix is simply a table showing the correlation coefficients between variables.

We can use the `corr()` function in Python to create a correlation matrix. We also use the `round()` function to round the output to two decimals:

```
In [11]: Corr_Matrix = round(full_health_data.corr(),2)
print(Corr_Matrix)

# Drop 2 columns - Hours_Work and Hours_Sleep to view the matrix nice.
#
health_part = full_health_data.drop(columns=['Hours_Work', 'Hours_Sleep'])
Corr_Matrix = round(health_part.corr(),2)
print(Corr_Matrix)
```

	Duration	Average_Pulse	Max_Pulse	Calorie_Burnage	\
Duration	1.00	-0.17	0.00	0.89	
Average_Pulse	-0.17	1.00	0.79	0.02	
Max_Pulse	0.00	0.79	1.00	0.20	
Calorie_Burnage	0.89	0.02	0.20	1.00	
Hours_Work	-0.12	-0.28	-0.27	-0.14	
Hours_Sleep	0.07	0.03	0.09	0.08	

	Hours_Work	Hours_Sleep
Duration	-0.12	0.07
Average_Pulse	-0.28	0.03
Max_Pulse	-0.27	0.09
Calorie_Burnage	-0.14	0.08
Hours_Work	1.00	-0.14
Hours_Sleep	-0.14	1.00

	Duration	Average_Pulse	Max_Pulse	Calorie_Burnage
Duration	1.00	-0.17	0.00	0.89
Average_Pulse	-0.17	1.00	0.79	0.02
Max_Pulse	0.00	0.79	1.00	0.20
Calorie_Burnage	0.89	0.02	0.20	1.00

Using a Heatmap

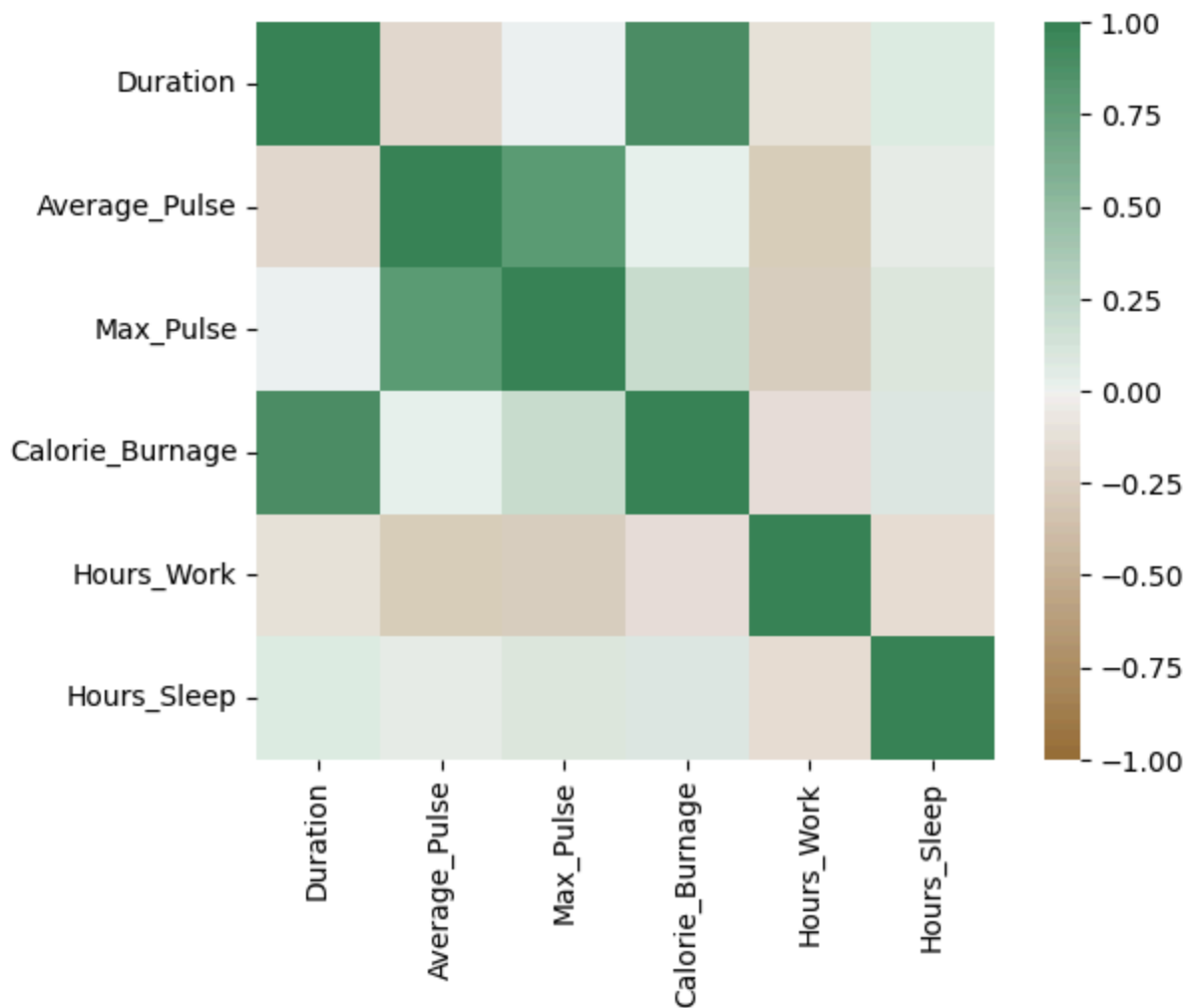
We can use a Heatmap to Visualize the Correlation Between Variables:

```
In [12]: import matplotlib.pyplot as plt
import seaborn as sns

correlation_full_health = full_health_data.corr()

axis_corr = sns.heatmap(
    correlation_full_health,
    vmin=-1, vmax=1, center=0,
    cmap=sns.diverging_palette(50, 500, n=500),
    square=True
)

plt.show()
```



1.6 Correlation Does not imply Causality

Correlation measures the numerical relationship between two variables.

A high correlation coefficient (close to 1), does not mean that we can for sure conclude an actual relationship between two variables.

A classic example:

- During the summer, the sale of ice cream at a beach increases
- Simultaneously, drowning accidents also increase as well

Question: Does this mean that increase of ice cream sale is a direct cause of increased drowning accidents?

1.7 Linear Regression

The term regression is used when you try to find the relationship between variables.

In Machine Learning and in statistical modeling, that relationship is used to predict the outcome of events.

We will use Scikit-learn to train various regression models. Scikit-learn is a popular Machine Learning (ML) library that offers various tools for creating and training ML algorithms, feature engineering, data cleaning, and evaluating and testing models. It was designed to be accessible, and to work seamlessly with popular libraries like NumPy and Pandas.

We see how to apply a simple regression model for predicting Calorie_Burnage on various factors such as Average_Pulse or Duration.

```
In [13]: # !pip install seaborn plotly

import numpy as np
import plotly.express as px
import plotly.graph_objects as go
from sklearn.linear_model import LinearRegression

df = full_health_data
X = df.Average_Pulse.values.reshape(-1, 1)

model = LinearRegression()
model.fit(X, df.Calorie_Burnage)

x_range = np.linspace(X.min(), X.max(), 100)
y_range = model.predict(x_range.reshape(-1, 1))

fig = px.scatter(df, x='Average_Pulse', y='Calorie_Burnage', opacity=0.65)
fig.add_traces(go.Scatter(x=x_range, y=y_range, name='Regression Fit'))
fig.show()
```

Question:

We have seen earlier how to apply a simple regression model for predicting Calorie_Burnage from Average_Pulse. There might be another candidate Duration in addition to Average_Pulse. You will need to repeat the above linear regression process to find relationship between Calorie_Burnage and Duration.

Comment on the both regression lines: Calorie_Burnage - Average_Pulse and Calorie_Burnage - Duration.

```
In [15]: import numpy as np
import plotly.express as px
import plotly.graph_objects as go
from sklearn.linear_model import LinearRegression

df = full_health_data
X = df.Duration.values.reshape(-1, 1)

model = LinearRegression()
model.fit(X, df.Calorie_Burnage)

x_range = np.linspace(X.min(), X.max(), 100)
y_range = model.predict(x_range.reshape(-1, 1))

fig = px.scatter(df, x='Duration', y='Calorie_Burnage', opacity=0.65)
fig.add_traces(go.Scatter(x=x_range, y=y_range, name='Regression Fit'))
fig.show()
```

SIT225 Data Capture Technologies

Assessment Pass Task 3P – Peer Review

Peer Review of Assessment 2 – Gordon

Reviewer: Deven Garg

Peer reviewed: Gordon

Date: 31 August 2026

The following feedback is based on the peer's Assessment 2 submission and addresses the eight review criteria specified in the SIT225 Task 3 peer review protocol.

Review Criterion	Peer Feedback
1. Use case clarity	The use case is clear and focuses on improving a low-cost anti-theft device by combining a PIR sensor with an LSM6DS3 accelerometer. The refined research question is more specific than the original problem because it focuses on reducing false alarms by combining different sensor information. The distinction between PIR detecting nearby human movement and the accelerometer detecting physical movement is easy to understand. However, the target of reducing false alarms by exactly 40% needs stronger justification because it is presented as a target without explaining how the value was selected. It would also help to define more clearly what counts as actual theft compared with normal activities such as typing, walking past the desk or accidentally touching the desk. Overall, the problem is researchable, but clearer operational definitions would make the study easier to reproduce.
2. Literature review rigour	The literature review includes two relevant anti-theft/security studies and identifies limitations in existing systems. It connects the literature to issues such as false alarms, sensor limitations and real-time monitoring. However, the literature base is relatively small, with only two main sources used to establish the research gap. Some claims about PIR false positives, accelerometer sensitivity and statistical requirements would benefit from direct citations to technical or peer-reviewed sources. The research gaps also move between vehicle security, smart homes and the proposed desk security device. Adding another source specifically related to multi-sensor fusion or PIR/accelerometer-based detection would strengthen the research justification.

3. Hypothesis quality	H1 is the strongest hypothesis because it provides measurable targets of greater than 95% theft detection and less than 5% false alarms when both sensors are used. The proposed 50 test runs also provide a practical way to compare the sensors. H2 is understandable because it compares short accidental bumps with longer movements associated with theft. However, there is an issue between H2 and the proposed 1 Hz sampling rate. If an accidental bump lasts for less than one second, a reading taken once per second could completely miss it. Gordon should either increase the sampling rate for this test or change H2 so that the events being investigated can reliably be measured at 1 Hz.
4. Sensor selection	The PIR sensor and LSM6DS3 accelerometer are appropriate choices because they measure different characteristics of a possible theft event. PIR detects nearby human movement while the accelerometer detects physical movement or shaking. Combining them is therefore useful for reducing false alarms. The use of acceleration magnitude from the X, Y and Z axes is also reasonable. However, the protocol could explain more clearly where the sensors are physically positioned and how their detection areas overlap. The reason why 1 Hz is sufficient should also be explained more clearly, particularly for short movements.
5. Ethical considerations	The ethical section is one of the stronger parts of the design because it recognises that motion information collected in a home can reveal people's routines even without names or other direct identifiers. The privacy notice, no-camera/no-microphone approach and defined retention periods are positive measures. However, the privacy notice refers to “facility managers” and “weekly summaries”, which does not seem to match the home-based testing described in the project. This should be changed to reflect the actual stakeholders. The consent process could also explain more clearly how family members can agree to participate and withdraw from testing.

6. Feasibility	The proposed experiment is generally feasible for a student because it uses relatively simple hardware, local CSV storage and a limited collection period. The 50 test runs are also manageable. However, the project includes several additional components such as Arduino Cloud monitoring, email/browser notifications, missing-data handling and statistical analysis. I recommend prioritising the core experiment first: comparing PIR alone, accelerometer alone and both sensors together. Cloud monitoring could be treated as an extension rather than being essential to validating the main hypotheses.
7. Strongest element	The strongest element is the decision to combine the PIR and accelerometer rather than relying on a single sensor. The two sensors observe different characteristics of an event, which gives the project a clear technical purpose. The proposed logic that an alert should require PIR movement and sustained item movement is a sensible approach to reducing false alarms. The project also has measurable performance targets rather than only demonstrating that the device works.
8. One improvement	The most important improvement would be to align the sampling rate, hypotheses and testing method. The current design uses 1 Hz sampling, but H2 claims that short bumps lasting less than one second can be distinguished from theft movements lasting several seconds. At one reading per second, a short bump could happen completely between two readings and never appear in the dataset. Increasing the sampling rate during controlled event testing, or redefining H2 around events that last long enough to be captured at 1 Hz, would make the hypothesis more scientifically defensible.

Acknowledgement

Reviewer: Deven Garg

Date: 31/08/2026

Part B: GenAI-Assisted Critique

Prompt 1: Protocol Design & Confounding Variables Critique

I have designed a sensor protocol for an anti-theft desk security system to differentiate between normal desk activities and actual theft while reducing false alarms. I am using a PIR motion sensor (body heat) and an LSM6DS3 accelerometer (3-axis motion/shaking) on an Arduino Nano 33 IoT board, sampling at 1 Hz for 24 hours. My hypotheses are:

1. *H1*: Combining PIR and accelerometer data reduces false alarm rates to <5% while maintaining >95% theft detection compared to single-sensor setups.
2. *H2*: Transient desk bumps last <1 second, whereas actual theft movement persists across consecutive readings (>2–3 seconds), which can be filtered out using a moving average / duration threshold.

What are potential weaknesses in this experimental design? What confounding environmental or physical variables (e.g., HVAC airflow, table movement, ambient vibration) might I have missed? Suggest 2–3 methodological improvements.

Prompt 2: Ethical & Privacy Gap Identification

I am deploying an anti-theft security device using PIR motion and accelerometer sensors in a home/office environment. My privacy protocol includes: logging non-biometric raw numbers (Timestamp, PIR_State, Accel_X/Y/Z, Accel_Delta) at 1 Hz, retaining raw CSVs locally for 7 days, aggregating data into daily summaries for up to 90 days, and displaying a physical privacy notice.

What ethical, power dynamic, or privacy gaps might I have overlooked regarding room occupants or family members? Who else could be affected by this data

collection? Suggest one additional safeguard to align with Privacy-by-Design principles (ISO 27001).

Prompt 3: Literature & Methodological Soundness

I identified a gap in current anti-theft research: existing smartphone/CCTV setups often produce high false-positive rates due to environmental noise or lack multi-sensor verification without continuous cloud dependence. My project tests Multi-Sensor Data Fusion (MSDF) combining PIR spatial detection and IMU acceleration magnitude at 1 Hz.

What statistical assumptions (e.g., temporal autocorrelation, idle noise normality) should I verify for 1 Hz time-series sensor data? Are there established data fusion frameworks or key papers in IoT physical security I should cite to justify my design?

Prompt one result Summary: Protocol Design & Confounding Variables Critique

Key Design Weaknesses

1. **Sampling Rate Too Slow (1 Hz):** Fast thefts (200–500 ms) will be missed or look identical to a table bump, breaking H_2 .
2. **PIR Sensor Blindspots:** HVAC/sunlight cause false positives; stationary thieves cause false negatives, threatening H_1 's >95% accuracy claim.
3. **Uncontrolled Confounders:** Fails to isolate signal from ambient vibration (footsteps/traffic), table resonance, thermal drift, and WiFi EMI.
4. **Vague "Theft" Definition:** Lacks precise physical metrics (speed, weight, distance) for what actually constitutes theft.

Key Methodological Fixes

- **Boost Sampling to 10–20+ Hz:** Enables frequency filtering (FFT) to easily separate quick bumps from sustained lifting.
- **Deploy a Sensor Array:** Uses multiple accelerometers across the desk to distinguish localized theft from whole-table vibrations.
- **48-Hour Stratified Baseline:** Logs environmental noise and HVAC cycles to create adaptive trigger thresholds.

Revised Hypotheses (In Short)

Refined H_1 : Multi-sensor fusion at 10+ Hz with spatial filtering drops false alarms <5% across all environmental cycles.

Refined H_2 : Theft produces localized, sustained (>0.5s) 1–3 Hz movement; bumps produce brief (<300 ms) broadband spikes.

Prompt Two result Summary: Ethical & Privacy Gap Identification

Key Privacy & Ethical Gaps

1. **Consent & Power Dynamics:** A static privacy notice lacks ongoing consent, withdrawal mechanisms, or protection for non-consenting occupants (e.g., family, guests, co-workers who feel pressured).
2. **Scope Creep & Inference Risk:** 1 Hz PIR and accelerometer logs reveal daily routines, presence, and absence patterns—risking secondary misuse like behavior or productivity tracking.
3. **Arbitrary Data Retention:** The 7-day raw / 90-day aggregate timeline lacks a documented data-minimization rationale; a shorter timeframe (e.g., 48 hours raw) would significantly reduce risk.
4. **Overlooked Stakeholders:** Guests, renters, family members, and third parties (e.g., landlords, insurers) could be impacted without direct consent.

Key Privacy-by-Design Safeguard

Role-Based Access Control (RBAC) & Audit Logging:

- **Device Owner:** Access to real-time alerts and short-term raw logs only.
- **Facility/Room Manager:** Access restricted strictly to high-level daily aggregates.
- **Audit Trail:** Log every access request (timestamp, user, purpose) to ensure accountability, purpose limitation, and transparency per **ISO 27001** standards.

Prompt Three result Summary: Literature & Methodological Soundness

Key Statistical Assumptions to Verify

1. **Temporal Autocorrelation:** 1 Hz samples are not independent. Run **ACF / Ljung-Box** tests to avoid inflating statistical significance.
2. **Stationarity & Noise:** Check for baseline drift (temperature/orientation) using **ADF tests**. Verify if idle noise is Gaussian (**Shapiro-Wilk**) to justify standard $\mu + 3\sigma$ thresholds.
3. **Sensor Lag & Independence:** Use **cross-correlation** to account for time lags between PIR heat detection and IMU shaking, and test for conditional independence before applying Naïve Bayes.

Critical Evaluation Metrics

- **Prioritize False Alarms:** Overall accuracy is misleading for security. Report **False Positive Rate (FPR)**, **False Alarms per Hour**, and **Detection Latency**.
- **Standard Classifiers:** Include **Precision**, **Recall**, **F1-Score**, and **ROC-AUC** to prove system reliability.

Fusion Frameworks & Core Literature

Theoretical Frameworks:

- **Dempster-Shafer Evidence Theory:** Best for handling sensor conflicts and uncertainty (e.g., PIR false triggers vs. accidental bumps).
- **Bayesian Data Fusion:** Provides lightweight, edge-friendly probabilistic decision-making.
- **JDL Model:** Frame your system as Feature-level or Decision-level fusion.

Key Citations to Include:

- **Hamda et al. (2023):** Justifies Dempster-Shafer theory for uncertainty in IoT multisensor fusion.
- **Yuan et al. (2022):** Supports PIR/multimodal fusion as a privacy-preserving alternative to continuous video.
- **Hall & Llinas:** Classic foundational citation for Multi-Sensor Data Fusion (MSDF).

Concise Refined Hypothesis

H1: A multi-sensor data fusion framework combining PIR motion detection and IMU acceleration magnitude at 1 Hz significantly reduces false-positive theft alerts compared with single-sensor approaches by requiring corroborating evidence from independent sensing modalities.

GenAI feedback summary

Prompt / Topic	GenAI Response (key points)	Tool used	Date accessed
Protocol design critique	1. Sampling Rate Too Slow (1 Hz): Fast thefts (200–500 ms) will be missed or look identical to a table bump, breaking H_2. 2. PIR Sensor Blindspots: HVAC/sunlight cause false positives; stationary thieves cause false negatives, threatening H_1's >95% accuracy claim. 3. Uncontrolled Confounders: Fails to isolate signal from ambient vibration (footsteps/traffic), table resonance, thermal drift, and WiFi EMI. 4. Vague "Theft" Definition: Lacks precise physical metrics (speed, weight, distance) for what actually constitutes theft.	Deakin GEM	August 2026
Ethical gaps	1. Consent & Power Dynamics: A static privacy notice lacks ongoing consent, withdrawal mechanisms, or protection for non-consenting occupants (e.g., family, guests, co-workers who feel pressured). 2. Scope Creep & Inference Risk: 1 Hz PIR and accelerometer logs reveal daily routines, presence, and absence patterns—risking secondary misuse like behavior or productivity tracking. 3. Arbitrary Data Retention: The 7-day raw / 90-day aggregate timeline lacks a documented data-minimization rationale; a shorter timeframe (e.g., 48 hours raw) would significantly reduce risk. 4. Overlooked Stakeholders: Guests, renters, family members, and third parties (e.g., landlords, insurers) could be impacted without	Deakin GEM	August 2026

	direct consent.		
Literature validation	1. Temporal Autocorrelation: 1 Hz samples are not independent. Run ACF / Ljung-Box tests to avoid inflating statistical significance. 2. Stationarity & Noise: Check for baseline drift (temperature/orientation) using ADF tests. Verify if idle noise is Gaussian (Shapiro-Wilk) to justify standard $\mu + 3\sigma$ thresholds. 3. Sensor Lag & Independence: Use cross-correlation to account for time lags between PIR heat detection and IMU shaking, and test for conditional independence before applying Naïve Bayes.	Copilot	August 2026

Prompt 1 Result: Protocol Design & Confounding Variables Critique

Claude Haiku 4.5: Fast responses with minimal wait time

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I have designed a sensor protocol for an anti-theft desk security system to differentiate between normal desk activities and actual theft while reducing false alarms. I am using a PIR motion sensor (body heat) and an LSM6DS3 accelerometer (3-axis motion/shaking) on an Arduino Nano 33 IoT board, sampling at 1 Hz for 24 hours. My hypotheses are:

1. H1: Combining PIR and accelerometer data reduces false alarm rates to <5% while maintaining >95% theft detection compared to single-sensor setups.

2. H2: Transient desk bumps last <1 second, whereas actual theft movement persists across consecutive readings (>2-3 seconds), which can be filtered out using a moving average / duration threshold.

What are potential weaknesses in this experimental design? What confounding environmental or physical variables (e.g., HVAC airflow, table movement, ambient vibration) might I have missed? Suggest 2-3 methodological improvements.

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Claude Haiku 4.5: Fast responses with minimal wait time

🔍 3 個來源

I'll evaluate your sensor protocol design. This is a solid starting point, but there are real weaknesses to address.

Weaknesses in Current Design

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Use of responses: Deakin GEM uses artificial intelligence to generate responses. While it aims to provide helpful and relevant information, responses may be incomplete, outdated, or inaccurate and should not be relied on as a sole source of information. Please review and verify information where appropriate.

Weaknesses in Current Design

1. Sampling rate too low for motion discrimination

The problem:

1 Hz (1 sample per second) is insufficient to distinguish a genuine theft event from a transient bump.

- A quick grab or lift might occupy 200–500 ms
- At 1 Hz, you may capture only 1–2 samples during the entire event
- You cannot reliably distinguish duration or acceleration *signature* at this resolution

Impact on H2:

Your threshold of >2–3 seconds requires consecutive readings, but with only 1 sample per second, you have very coarse temporal resolution. A thief grabbing something in 0.5 seconds might appear identical to a bump.

2. PIR sensor limitations underexplored

- **False positives:** HVAC drafts, sunlight changes, and radiator heat can trigger PIR
- **False negatives:** A person standing still (common during theft preparation) won't trigger a body-heat sensor
- **Assumption mismatch:** You're assuming theft *always involves motion*; it often doesn't (person waits, then grabs)

This directly threatens H1's claim of >95% detection.

3. Confounding variables not controlled

Your design doesn't systematically isolate signal from noise:

FACTOR	EFFECT	YOUR CONTROL?	 
HVAC airflow	Pushes lightweight objects; triggers PIR false positives	✗ None mentioned	
Ambient vibration (footsteps, traffic, doors)	Accelerometer noise floor	✗ Not baseline-tested	
Table resonance	A bump on one desk corner radiates to another; accelerometer picks it up globally	✗ Single sensor placement	
Temperature drift	PIR baseline shifts over 24h; sensitivity drifts	✗ Not corrected	
Electromagnetic interference	LSM6DS3 and Arduino on same board; EMI from WiFi (IoT)	✗ No shielding/filtering mentioned	

4. Weak operational definition of "theft"

What counts as theft in your test?

- How fast must an object move?
- What mass?
- Does the object leave the desk, or just get relocated?

Without this, H1 and H2 become unfalsifiable.

Methodological Improvements

Improvement 1: Increase sampling rate + add spectral filtering

Change:

- Sample accelerometer at **10–20 Hz** (minimum; 50 Hz preferred)
- Keep PIR at 1 Hz if power-constrained, but timestamp it precisely

Why:



Methodological Improvements

Improvement 1: Increase sampling rate + add spectral filtering

Change:

- Sample accelerometer at **10–20 Hz** (minimum; 50 Hz preferred)
- Keep PIR at 1 Hz if power-constrained, but timestamp it precisely

Why:

- You can now resolve acceleration shape: a bump is a sharp spike; theft is typically a sustained ramp or multiple peaks
- Compute **frequency domain features** (FFT) on 2–5 second windows:
 - Theft movements often cluster in 1–3 Hz band (deliberate lifting)
 - Environmental vibration typically >5 Hz (traffic, HVAC harmonics)
- Apply a high-pass filter (>0.5 Hz) to remove drift; notch filter at 50/60 Hz for power-line coupling

Outcome: Separates genuine motion signatures from environmental noise.

Improvement 2: Multi-location accelerometer array + spatial filtering

Change:

- Deploy **3 accelerometers** at different desk locations (corner, center, opposite corner)
- Cross-correlate signals across sensors

Why:

- **Local vs. global motion:** A thief lifting an object from corner A shows high local acceleration there but coordinated motion across sensors
- A bump or vibration affecting the whole desk shows *synchronized* acceleration everywhere
- Spatial correlation distinguishes **localized force** (theft) from **distributed disturbance** (HVAC, footsteps)

Outcome: Reduces false positives from ambient vibration; improves specificity of H1.

Improvement 3: Controlled baseline + environmental stratification

Change:

- Before testing, collect **48-hour baseline** data under controlled conditions:
 - Normal office activity (no theft attempts)
 - HVAC on/off cycles measured separately
 - Calibrated thermal drift (PIR sensitivity over time)
- Build a **per-condition model**:
 - Morning (cool, minimal HVAC)
 - Afternoon (heated, active HVAC)
 - Night (stable)

Why:

- Confounds (temperature, ambient vibration) are not *random*—they're systematic and predictable
- By stratifying, you can correct for time-of-day effects and seasonal HVAC patterns
- Your thresholds (duration, amplitude) become **adaptive**, not fixed—much more robust

Outcome: Eliminates systematic bias; makes H1 and H2 testable against real confounds.

Revised Hypotheses (Recommended)

Stronger H1:

"Multi-sensor fusion (PIR + 3-axis accelerometer at 10+ Hz) + spatial filtering reduces false alarm rates below 5% on controlled test data compared to single-sensor baselines, when baseline drift and HVAC cycles are accounted for."

Stronger H2:

"Theft events produce sustained acceleration signatures (>0.5 s) in the 1–3 Hz band localized to one desk zone; bumps produce brief (<300 ms) broadband transients. A 2-second moving-average threshold with spatial filtering correctly classifies 90%+ of events."

Quick Checklist

- ☐ Increase accelerometer sampling to ≥ 10 Hz
- ☐ Characterise PIR + accelerometer baseline drift over 48 h
- ☐ Define operational theft protocol (speed, mass, object type)
- ☐ Add spatial accelerometer array or at least rotate sensor placement
- ☐ Test HVAC on/off cycles as separate conditions
- ☐ Compute frequency features (FFT) in addition to amplitude
- ☐ Report false positive and false negative rates separately

This is a realistic system. The tweaks above move you from exploratory testing to a defensible protocol.

Prompt 2 Result: Ethical & Privacy Gap Identification

Deakin GEM | Students

Desk Anti-Theft Sensor Des...

https://gem.deakin.edu.au/c/882319f8-4f28-4547-afb0-bfa83060b19

Claude Haiku 4.5: Fast responses with minimal wait time

I am deploying an anti-theft security device using PIR motion and accelerometer sensors in a home/office environment. My privacy protocol includes: logging non-biometric raw numbers (Timestamp, PIR_State, Accel_X/Y/Z, Accel_Delta) at 1 Hz, retaining raw CSVs locally for 7 days, aggregating data into daily summaries for up to 90 days, and displaying a physical privacy notice.

What ethical, power dynamic, or privacy gaps might I have overlooked regarding room occupants or family members? Who else could be affected by this data collection? Suggest one additional safeguard to align with Privacy-by-Design principles (ISO 27001).

Claude Haiku 4.5: Fast responses with minimal wait time

搜尋到 2 個來源

I need to address this directly: **the provided context is about a LiteLLM/AWS Bedrock bug and is not relevant to your privacy question.** I'm setting it aside and answering from first principles and general privacy knowledge.

Privacy Gaps in Your Protocol

1. Consent and awareness asymmetry

Your physical privacy notice informs people that data is collected, but:

- **Who consents?** If this is a shared home/office, does every occupant have equal say, or only the owner/manager?
- **Ongoing consent:** A notice posted once doesn't capti evolving comfort levels or new occupants who didn't see it initially

傳送訊息

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- **Ongoing consent:** A notice posted once doesn't capture *evolving* comfort levels or new occupants who didn't see it initially
- **Power imbalance:** Employees or family members may feel unable to object without social/professional cost

Gap: No mechanism for withdrawing consent or opting out of specific rooms.

2. Scope creep and secondary use

You've defined the current protocol, but:

- Who controls aggregated daily summaries after 90 days? Can they be shared with insurers, landlords, or authorities?
- Could accelerometer data (e.g., patterns of "person standing still at desk for 8 hours") be reused for productivity tracking or behaviour profiling?
- Is there contractual clarity on data deletion after 90 days, or does it persist indefinitely?

Gap: No explicit secondary-use restrictions or data governance policy.

3. Room-level granularity and inference risk

You're not logging *who* is present, but:

- Accelerometer + PIR patterns can infer room occupancy, movement habits, and even *absence patterns* (e.g., "bedroom unoccupied 11 pm–7 am")
- If multiple people share a space, you cannot distinguish them—but aggregated motion patterns still reveal *something* about collective behaviour
- In a family home, this indirectly profiles children, guests, or vulnerable persons

Gap: No de-identification or aggregation safeguard for multi-occupant spaces.

4. Data retention justification

Why 7 days raw + 90 days aggregated?

- Is this driven by theft investigation timelines, or just "seems reasonable"?
- Shorter retention (e.g., 48 hours raw, 30 days summary) would reduce risk

Gap: No documented data minimisation rationale.

Who Else Is Affected?

- **Occupants who didn't consent** (renters, employees, family members)
 - **Guests** (unaware they're being monitored)
 - **Third parties** receiving aggregated data (insurers, property managers, law enforcement if subpoenaed)
 - **Vulnerable groups** (children, elderly, unwell persons whose movement patterns reveal health/safety information)
-

Recommended Additional Safeguard (Privacy-by-Design)

Implement role-based data access with audit logging

Recommended Additional Safeguard (Privacy-by-Design)

Implement role-based data access with audit logging

What:

- Define explicit roles: *device owner, facility manager, security investigator*
- Each role can access only the minimum required:
 - **Owner:** Real-time alerts + 7-day raw logs (if theft suspected)
 - **Facility manager:** Daily aggregates (occupancy trends only)
 - **Investigator:** Full 90-day history (only with documented consent and specific incident)
- Log every access attempt with timestamp, requester, and purpose

Why this aligns with ISO 27001 & Privacy-by-Design:

- **Purpose limitation:** Access tied to defined roles, not open access
- **Accountability:** Audit trail proves who accessed what, when, why
- **Principle of least privilege:** Each user sees only necessary data
- **Transparency:** Occupants can request access logs to see who's viewed *their* data

Implementation:

🔍 搜尋 儲存 複製

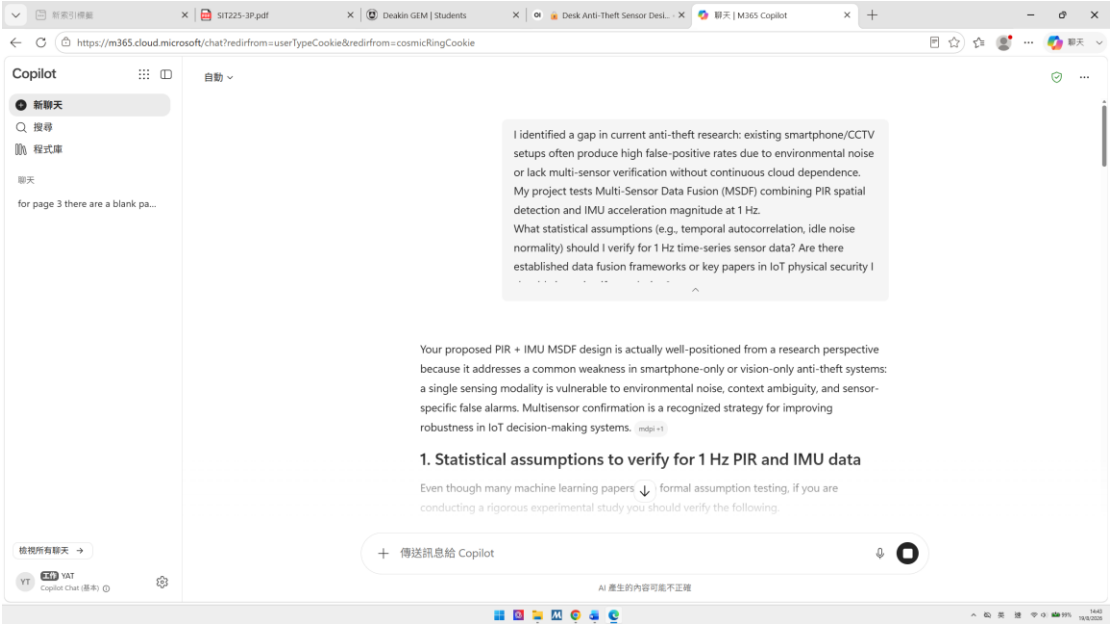
1	Access_Log: [Timestamp, User, Role, Data_Type, Purpose, Approved_By]
2	Example: 2026-08-19 14:30 alice@home Owner 7d_raw theft_incident self

Quick Checklist

- ☐ Document consent mechanism (how do occupants opt out or object?)
- ☐ Define and publish secondary-use policy (what happens to 90-day summaries?)
- ☐ Shorten retention to minimum viable (48 h raw, 30 d aggregate?)
- ☐ Add role-based access control + audit logging
- ☐ Test inference risk: can movement patterns alone reveal health/behaviour?
- ☐ Provide occupants a way to view/request deletion of their data

This shifts you from *notification* to *control*, which is Privacy-by-Design.

Prompt 3 Result: Literature & Methodological Soundness



Your proposed PIR + IMU MSDF design is actually well-positioned from a research perspective because it addresses a common weakness in smartphone-only or vision-only anti-theft systems: a single sensing modality is vulnerable to environmental noise, context ambiguity, and sensor-specific false alarms. Multisensor confirmation is a recognized strategy for improving robustness in IoT decision-making systems. mdpi +1

1. Statistical assumptions to verify for 1 Hz PIR and IMU data

Even though many machine learning papers skip formal assumption testing, if you are conducting a rigorous experimental study you should verify the following.

A. Temporal autocorrelation

At 1 Hz, consecutive observations are rarely independent.

Examples:

- PIR detections often remain active for several consecutive seconds after motion.
- IMU acceleration magnitudes show persistence during movement episodes.

Tests:

- Autocorrelation Function (ACF)
- Partial Autocorrelation Function (PACF)
- Durbin-Watson statistic
- Ljung-Box test

Why important:

If strong autocorrelation exists, treating each second as an independent sample will inflate significance tests and underestimate variance.

B. Stationarity

Security sensors often exhibit changing baselines:

- PIR sensitivity changes with ambient temperature.
- Smartphone IMUs experience orientation drift.
- Environmental conditions vary over time.

Tests:

- Augmented Dickey-Fuller (ADF)

B. Stationarity

Security sensors often exhibit changing baselines:

- PIR sensitivity changes with ambient temperature.
- Smartphone IMUs experience orientation drift.
- Environmental conditions vary over time.

Tests:

- Augmented Dickey-Fuller (ADF)
- KPSS

If non-stationary:

- Difference the series
 - Use rolling baselines
 - Normalize per observation window
-

C. Idle-state noise distribution

For your "device not being stolen" periods, characterize sensor noise.

For IMU magnitude:

$$a_t = \sqrt{x_t^2 + y_t^2 + z_t^2}$$

Check whether idle-state residuals follow:

- Gaussian distribution
- Laplace distribution
- Heavy-tailed distribution

Tests:

- Shapiro-Wilk
- Anderson-Darling
- Q-Q plots

Why useful:

Why useful:

Many threshold-setting methods assume normality.

For example:

$$Threshold = \mu + 3\sigma$$

is only justified when noise is approximately Gaussian.

D. Homoscedasticity (constant variance)

Variance may differ between:

- Day/night
- Indoor/outdoor
- Different device placements

Tests:

- Levene's test
- Bartlett's test

If variance changes substantially, adaptive thresholds are preferable.

E. Sensor synchronization

A key assumption in fusion systems is that PIR and IMU events occur within a meaningful temporal window.

Verify:

- Cross-correlation
- Lag analysis

Example:

If PIR activates 1-2 s before significant IMU acceleration, a fusion rule should account for this lag instead of requiring exact simultaneity.



F. Conditional independence

This is especially important if you use Bayesian fusion.

The naïve assumption is:

$$P(PIR, IMU|Theft) = P(PIR|Theft) \times P(IMU|Theft)$$

In practice, PIR and IMU may become correlated during theft events.

You can evaluate:

- Mutual information
- Pearson/Spearman correlation
- Contingency tables

High dependence suggests more advanced fusion models.

2. Performance metrics you should report

Because your project claims reduced false positives, these metrics are more important than overall accuracy:

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = \frac{2PR}{P + R}$$

Also report:

- False Positive Rate (FPR)
- False Alarm Rate per hour
- Detection latency (seconds)
- ROC-AUC if using probabilistic fusion

For security systems, false alarms are often more important than raw accuracy.

3. Established sensor fusion frameworks you can cite

JDL Data Fusion Model

The Joint Directors of Laboratories (JDL) model is one of the classic sensor-fusion architectures.

It separates fusion into levels:

- Signal/data level
- Feature level
- Decision level

Your PIR + IMU approach naturally fits a feature-level or decision-level fusion architecture. [cite](#)

Bayesian Data Fusion

Good choice when:

- Sensor reliabilities can be estimated.
- Probabilistic outputs are available.

A simple formulation:

$$P(Theft|PIR, IMU) = \frac{P(PIR, IMU|Theft)P(Theft)}{P(PIR, IMU)}$$

This is probably the strongest theoretical justification for a lightweight edge-based anti-theft system.

Dempster-Shafer Evidence Theory

Particularly useful when sensors are uncertain or occasionally conflict.

Advantages:

- Explicitly models uncertainty.
- Handles incomplete information.
- Widely applied in IoT multisensor fusion research. [model + 1](#)

This may fit your work very well because:



- PIR can generate false activations.

- Explicitly models uncertainty.
- Handles incomplete information.
- Widely applied in IoT multisensor fusion research. indpi +1

This may fit your work very well because:

- PIR can generate false activations.
- IMU can experience accidental movement.
- Fusion can combine confidence from both sources.

Kalman Filtering

Kalman filters are the classical approach for continuous-valued sensor fusion and state estimation. They are commonly cited as a standard baseline in multisensor systems. crdn +1

For a 1 Hz anti-theft prototype, however, Bayesian or Dempster-Shafer fusion may be easier to justify than a full state-space model.

4. Relevant papers directly supporting your design

I would strongly recommend citing:

(1) Hamda et al., 2023

"Multisensor Data Fusion in IoT Environments in Dempster-Shafer Theory Setting"

Contributions:

- Discusses uncertainty management in IoT fusion systems.
- Demonstrates improved decision accuracy using evidence-based fusion.
- Provides a strong theoretical rationale for combining heterogeneous sensors. indpi +1

(2) Yuan et al., 2022

"Interpretable Passive Multi-Modal Sensor Fusion for Human Identification and Activity Recognition"

Important because:

- Uses PIR within a multimodal sensing system.
- Shows sensor fusion improves robustness versus single-sensor solutions.
- Emphasizes privacy-preserving alternatives to camera-based monitoring. indpi +1

This is especially relevant if you argue that PIR+IMU reduces dependence on continuous CCTV/cloud processing.

(3) Hall & Llinas

"An Introduction to Multisensor Data Fusion"

A foundational citation commonly used to justify:

- Reduced uncertainty
- Increased reliability
- Better decision-making through complementary sensors

This paper underpins much of the modern JDL fusion literature.

5. A concise research hypothesis

Your methodology could be justified with:

HT: A multi-sensor data fusion framework combining PIR motion detection and smartphone IMU acceleration magnitude at 1 Hz significantly reduces false-positive theft alerts compared with single-sensor detection approaches by requiring corroborating evidence from independent sensing modalities.

That hypothesis aligns well with the established literature on complementary sensor fusion, uncertainty reduction, and decision-level IoT security fusion. indpi +1

For a student research paper, the strongest statistical section would include: **ACF/Ljung-Box** (autocorrelation), **ADF** (stationarity), **Shapiro-Wilk** (idle-noise normality), **cross-correlation** (sensor lag), and **confusion-matrix-based false-positive analysis**. Those tests directly support the validity of a 1 Hz PIR-IMU fusion design.

Part C: Critical Response Report

Introduction

This report discusses the feedback I received from my peer reviewer, Deven Garg, and two AI tools, Deakin GEM and Microsoft Copilot, about my security system. My original design uses an Arduino board, a PIR motion sensor and an LSM6DS3 accelerometer to detect possible desk theft. My aim was to detect more than 95% of theft events while keeping false alarms below 5%. The sensors were originally set to collect data once every second (1 Hz).

The feedback helped me find some problems with my original testing plan. In particular, I needed to improve the sensor reading speed, deal with environmental noise, improve my privacy plan and make sure my data analysis was suitable for the type of data I collected.

Feedback Summary & Thematic Analysis

Theme	Peer feedback	AI feedback	Consensus?
Sampling Rate	pointed out that 1 Hz could miss quick desk bumps that last less than one second.	reading once per second could miss short movements and suggested increasing the rate to around 10–20 times per second.	yes, Both agree 1 Hz is too slow.
Unwanted Noise & Confounders	things such as heat drafts and table vibrations could cause false alarms if they are not controlled.	AI feedback mentioned sunlight, air conditioners and floor vibrations as possible sources of unwanted	Yes, Both agree environmental noise controls are missing.

		sensor readings. It also suggested collecting baseline data first. first.	
Privacy & Ethics	my original design mentioned “facility managers”, which does not really fit a personal desk setup. He also suggested making consent clearer.	movement data could still show information about a person's daily routine, even if no names are recorded.	Yes, Both agree that the privacy section needs changes.
Math & Literature Support	suggested adding more scientific research about security sensors.	checking the statistical methods because sensor readings collected over time may be related to each other. It also suggested using research about multi-sensor systems.	Yes, Both agree that more research and statistical support would be useful.

Four main issues came out of the feedback:

- **Slow Reading Speed:** Both Deven and AI said that 1 Hz is too slow for short events. For example, a quick bump lasting 0.2 seconds could happen between two readings and be missed.
- **Environmental Noise:** Things such as temperature changes, air conditioning and normal vibrations could affect the sensors and cause false alarms.
- **Privacy Problems:** “facility managers” did not fit my home desk situation. The AI feedback also made me think more about how movement timestamps could show when someone is at their desk or away.
- **Statistical Testing:** AI pointed out that readings taken close together in time may not be completely independent. So might need to check this before using some statistical methods.

Identify Conflicts or Tensions

There was some difference between the feedback about how much I should increase the sampling rate.

AI suggested increasing the sensor reading rate to between 10 and 50 times per second. However, I also need to consider the time and hardware limits.

My solution: I think I should increase the accelerometer to 10 Hz on the Arduino. I will still summaries the results every second so that I do not create unnecessarily large amounts of stored data.

Critical Evaluation: Accept / Reject / Revise

Feedback Point	Your decision (Accept / Reject / Revise)	Justification with evidence	Literature / source
Increase reading speed from 1 Hz to 10 Hz	Accept	At 1 Hz, short bumps could be missed. Increasing it to 10 Hz gives the	Deakin GEM AI, Deven Garg

		system more chances to record quick movements.	
Fix heat and draft false alarms	Revise	PIR sensors may react to changes in the environment. But I am not sure how sensitive for PIR to detect environment such as sunlight. Therefore I might need to test under different environment.	
Shorten data storage time	Accept	Keeping raw sensor data for 7 days is more than I need. I will delete the raw data after 48 hours to reduce privacy risks.	ISO 27001, Deakin GEM
Fix wrong user roles	Accept	I removed references to “facility managers” because the system is designed for a personal desk rather than a facility.	Deven Garg
Add statistical independence tests	Accept	Sensor readings collected over time can be	Microsoft Copilot

		related to each other. I will use checks such as ACF and the Ljung-Box test before interpreting the data.	
Add 3 accelerometers across the desk	Reject	Three sensors would add more wiring, testing and setup work than I can reasonably manage. Also don't have enough budget to buy more sensor.	Feasibility limits

1. Faster Sensor Reading Speed (Accept)

Both my peer reviewer and the AI feedback pointed out that 1 Hz is too slow for my experiment. A desk bump can happen very quickly, possibly in less than 0.3 seconds. If the sensor only takes one reading every second, there is a chance that the movement will not be recorded properly.

Because of this, I changed the accelerometer sampling rate from 1 Hz to 10 Hz. This means the board can collect more readings during a short movement and gives me more information to work with.

2. Controlling Environmental Noise (Revise)

The feedback also made me think more about environmental factors. Changes caused by sunlight, heaters or air conditioning could affect the PIR sensor, while the accelerometer may also pick up normal vibrations.

To deal with this, I will collect baseline data before starting the main experiment. This will give me an idea of the normal sensor readings in the room. I can then use this information when setting the alarm thresholds.

3. Better Data Privacy (Accept)

Deven pointed out that my original privacy section included “facility managers”, which does not fit my project because it is designed for a personal. I changed the roles to Main Desk User and Room Occupants.

I also changed how long data is kept. Raw sensor data will now be deleted after 48 hours instead of 7 days. Daily summaries will be kept for 30 days instead of 90 days. This reduces the amount of information stored about people's movements.

4. Math Tests and Sensor Array (Accept & Reject)

Copilot pointed out that sensor readings collected one after another may be related to previous readings. I accepted the suggestion to check this using statistical tests such as autocorrelation.

However, I decided not to use three separate accelerometers. Although this could provide more information about movement across the desk, it would also make the hardware setup and testing more complicated.

Protocol Revision Summary

Original Plan

- **Sensor Reading:** 1 Hz (once per second) for all sensors.
- **Hypotheses:** I originally thought 1 Hz data would be enough to distinguish short bumps from theft using simple averages.
- **Privacy:** Raw data was stored for 7 days and summaries for 90 days. The design also included “facility managers”, which did not fit the intended setup.

Change:

- **Sensor Reading:** The accelerometer will now run at 10 Hz so that quick movements are less likely to be missed. The PIR sensor will remain at 1 Hz.

New Hypotheses:

H1: Combining PIR motion data with 10 Hz accelerometer data will keep false alarms below 5% while maintaining detection above 95%.

H2: Bumps will produce short spikes lasting less than 0.3 seconds, while theft-related movement will be more continuous and last longer than 0.5 seconds. The 10 Hz sampling rate should make this difference easier to identify.

- **Privacy Updates:** The stakeholders are now the desk user and other people sharing the room, such as housemates or coworkers. Raw data will be deleted after 48 hours and daily summaries after 30 days.
- **Analysis Updates:** statistical checks for the time-series data and added research to support the use of multiple sensors.

Reflection: Evaluative Judgement & Professional Growth

The feedback from my peer and the AI tools helped me find problems that I had not noticed in my original design.

At first, I thought 1 Hz was enough because it would reduce memory and power use. After looking at the feedback, I realised that this could actually make it difficult to detect short bumps. I changed the accelerometer to 10 Hz so that more of these quick movements can be recorded.

I also decided not to follow the suggestion of using three accelerometers. Although having more sensors could give me more information, it would make the project harder to build and test within the time limit. This helped me understand that a good research idea also needs to be realistic for the available time and hardware.

The privacy feedback was also useful. I originally focused mainly on names and personal information, but I realised that sensor timestamps can also tell us something about a person's routine. For example, they could show when someone is usually at their desk or away from it. Because of this, I reduced the amount of time that sensor data is stored and changed the user roles in my design. This made me think more carefully about privacy when designing a real-world system.

Acknowledgement of GenAI Use

Per Deakin policy, include an Acknowledgements section:

"This response report was developed with assistance from the following approved Deakin genAI tools:

- Deakin GEM (accessed September 2026): Used to generate structured critique on protocol design and confounding variables (3 iterations).
- Microsoft Copilot for web (accessed September 2026): Used to summarise ASHRAE comfort standards and identify relevant literature.

All genAI-generated suggestions were critically evaluated and integrated only where supported by evidence or peer consensus. The final design, decisions, and justifications are my own."